LAB - 5

DATA FLOW DIAGRAM

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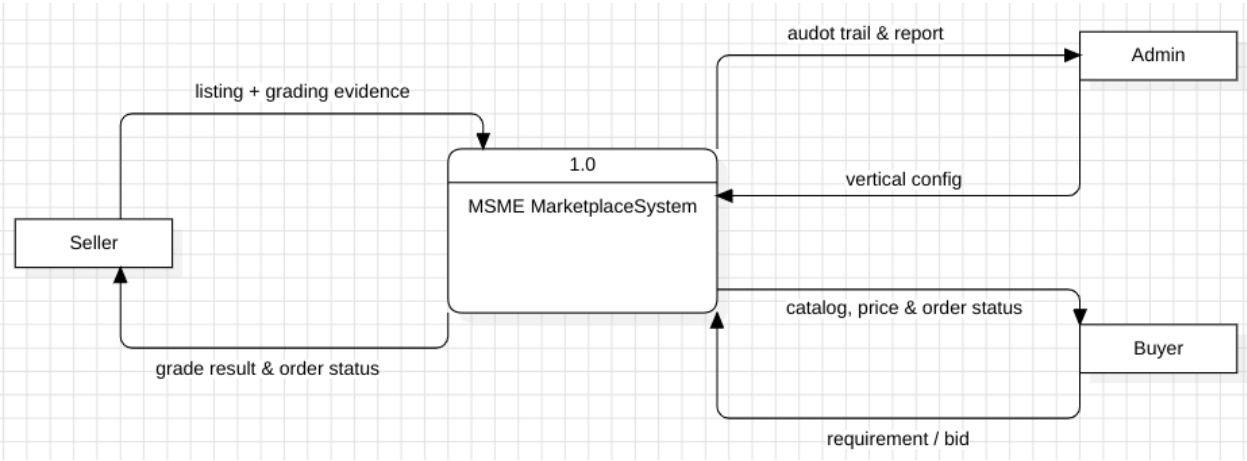
Level 0: Context Diagram

- **Purpose:** Provides a high-level overview of the MSME Multi-Vertical Commodity Marketplace System.
- **Main Entities:**

  Seller: Submits commodity listings and grading evidence to the system. Receives grade results and order status updates.

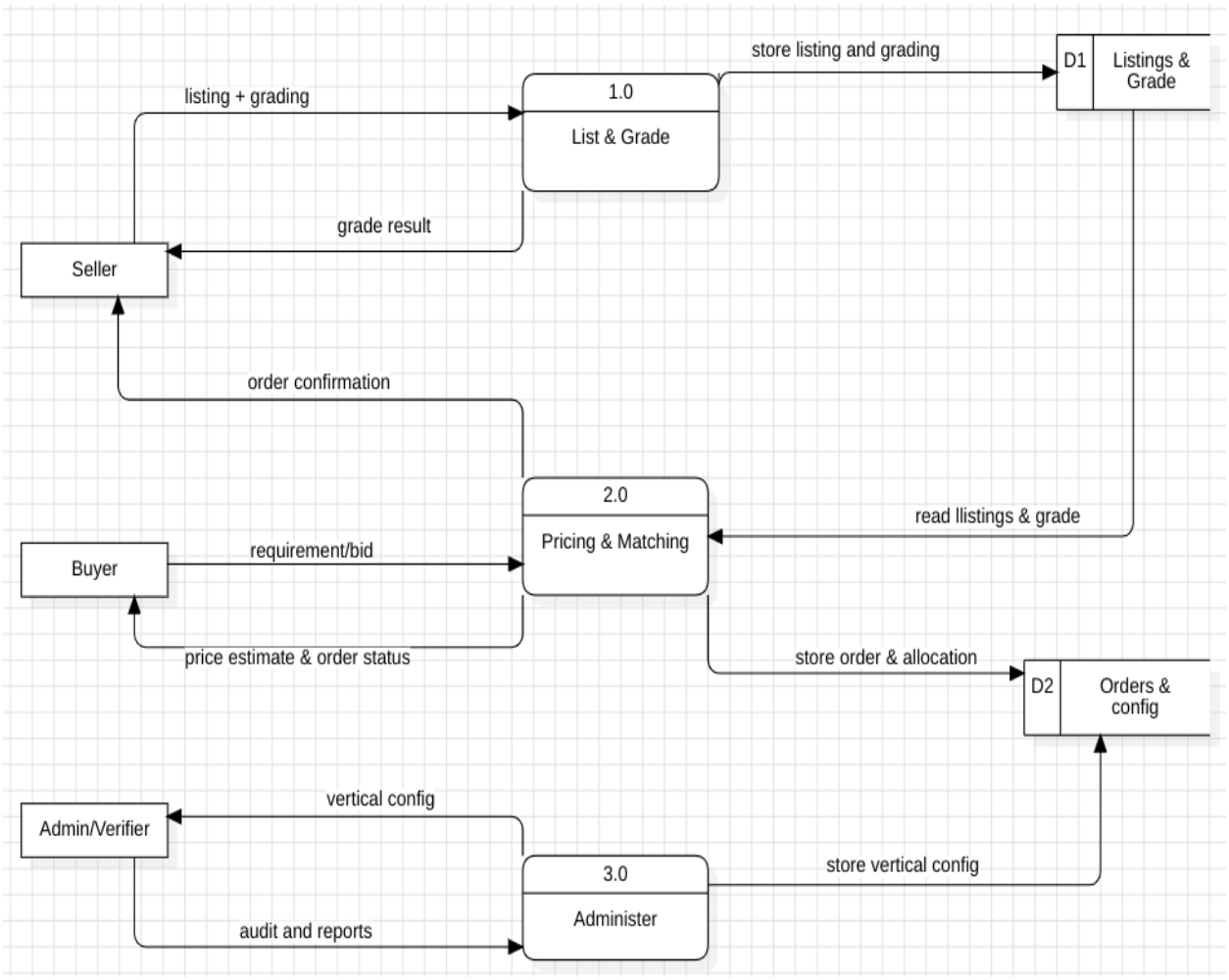
  Buyer: Posts requirements or bids to the marketplace. Receives catalog information, estimated prices, and finalized order status.
- **Secondary Entities:**

  Admin: Configures plug-in models for target verticals. Receives audit trails and reports for platform administration and dispute resolution.
- **System Representation:** The MSME Marketplace System is represented as a single main process (Process 1.0), with internal matching engines, grading logic.



Level 1: MSME Marketplace System

- **Purpose:** Decomposes the MSME Marketplace System into its major functional processes and shows data movement.
- **External Entities:**
  Seller
  Buyer
  Admin/Verifier
- **Main Processes:**
  1.0 List & Grade: Accepts the listing details and grading evidence from the Seller. Processes the grading request and returns the resulting grade back to the Seller.
  2.0 Pricing & Matching: Takes requirements and bids from the Buyer. It reads existing listings to execute order allocation, returning price estimates and order statuses to the Buyer, while sending order confirmations to the Seller.
  3.0 Administer: Receives the vertical configuration parameters inputted by the Admin/Verifier. It generates the necessary audit trails and reports for administrative review.
- **Data Stores:**
  D1 - Listings & Grade: Stores the sellers' active listing data alongside their confirmed grading results generated by process 1.0.
  D2 - Orders & config: Stores the finalized orders and allocation data generated from process 2.0, as well as the vertical configuration schema defined in process 3.0.


store listing and grading
D1
Listings & Grade
listing + grading
1.0
List & Grade
grade result
Seller
order confirmation
2.0
Pricing & Matching
read llistings & grade
Buyer
requirement/bid
price estimate & order status
store order & allocation
D2
Orders & config
vertical config
Admin/Verifier
3.0
Administer
store vertical config
audit and reports

Level 2: Detailed MSME Marketplace System Processes

- **Purpose:** Provides a detailed decomposition of the major functional processes (1.0, 2.0, and 3.0) from the Level 1 DFD to illustrate granular data movement and internal sub-processes on a single diagram.
- **External Entities:**

  Seller: Submits commodity listings and grading evidence, and receives grade results and finalized order confirmations.

  Buyer: Posts commodity requirements or bids, and receives price estimates and order status updates.

  Admin/Verifier: Inputs vertical configurations and dispute details, and receives system audit reports.
- **Detailed Sub-Processes:**

  1.1 Capture Listing Details: Receives the initial commodity specifications and details from the seller.

  1.2 Evaluate Grade: Analyzes the uploaded evidence with the vertical's schema using an AI model or a human verifier.

  1.3 Finalize & Store Listing: Combines the captured listing details with the verified grade to format it for database storage.

  2.1 Calculate Price Estimate: Computes a cost estimate based on market feeds, verified grades in response to buyer requirement.

  2.2 Match & Allocate: Reads available active listings and matches them against buyer criteria to fulfill orders.

  2.3 Process Bid & Finalize Order: Handles negotiation mechanisms, finalizes the matched transaction, and distributes order statuses.

  3.1 Define Vertical Configuration: Processes and establishes the plug-in rules for acceptable units, grading schemas, and price calculations.

  3.2 Manage Disputes: Processes flagged workflows regarding grading accuracy or order fulfillment issues.

  3.3 Generate Audit Reports: Compiles historical grading decisions, timestamps, and order data into comprehensive reports.

- **Data Stores:**

  D1 - Listings & Grade: Stores the seller's active listings alongside their verified grading results.

  D2 - Orders & config: Stores the defined vertical configuration schemas, dispute resolution statuses, and final allocated orders

